00-H-581a
March 20, 1968
SUPERSEDING
Fed. Spec. 00-H-581
May 20, 1955
(See 6.11)

FEDERAL SPECIFICATION

HOLDERS, CUTTING TOOL; ENGINE LATHE, SHAPER AND PLANER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single point engine lathe, shaper and planer tool holders and knurling tools.

1.1.1 Federal specification coverage. Federal specifications do not include all types, classes, and styles of the commodity indicated by the titles of the specifications, or which are commercially available, but are intended to cover those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Tool holders covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I. Holder, lathe turning (broached hole or bit channel parallel to base).
- Class 1. Straight shank (single size).
- Class 2. Left-hand offset shank.
  - Style A. Rigid (single size).
  - Style B. Multi-size tool.
- Class 3. Right-hand offset shank.
  - Style A. Rigid (single size).
  - Style B. Multi-size tool.

Type II. Holder, lathe, turning (14° tilted broached hole or bit channel).
- Class 1. Straight shank, rigid (single size).
- Class 2. Left-hand offset shank.
  - Style A. Rigid (single size).
  - Style B. Multi-size tool.
- Class 3. Right-hand offset shank.
  - Style A. Rigid (single size).
  - Style B. Multi-size tool.

Type III. Holder, lathe, cut-off or side cutting.
- Class 1. Straight shank.
  - Style A. Rigid (single size).
  - Style B. Spring.
- Class 2. Left-hand offset shank.
  - Style A. Rigid (single size).
  - Style B. Spring.
- Class 3. Right-hand offset shank.
  - Style A. Rigid (single size).
  - Style B. Spring.

Type IV. Holder, lathe, threading.
- Class 1. Rigid.
- Class 2. Spring style.

Type V. Holder, lathe, knurling.
- Class 1. Single.
- Class 2. Multiple.

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Type VI. Holder, lathe, boring.
Class 1. Shank mounted.
Style A. Reversible, boring and turning.
Style B. Clamp block.
Style C. Split clamp.
Class 2. T-slot mounted.
Style A. Adjustable height.
Style B. Fixed height.
Type VII. Holder, planer and shaper, adjustable.
Type VIII. Holder, shaper, extension.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

00-C-851 - Cutter Bit, Tool Holder (Solid High-Speed Steel).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:
H28 - Screw-Thread-Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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United States of America Standards Institute (USASI) Handbook:
B94.6 - Knurling.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N.Y. 10016.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Tool holders and components shall be of the best quality of material as hereinafter specified.

3.1.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of holders which are otherwise in accordance with the requirements of this specification.

3.2 Design. Tool holders shall be so designed as to satisfactorily perform the functions of the types and classes as hereinafter specified. They shall be sufficiently strong to support tools rigidly when machining at maximum tool load. All external edges not necessarily required to be sharp for the utility of the tool, shall be rounded, beveled, or relieved in accordance with the best commercial practice. Where bolts are used as a clamping device, the thread shall be class 2A fit in accordance with H28.

3.3 Wrenches. Each tool holder and boring bar shall be supplied with an applicable wrench for use in clamping the tool bit to the tool holder and clamping the tool bit to the boring bar. Wrenches shall be of steel, heat-treated and of a suitable size, shape, and finish to insure proper performance.

3.4 Finish. Unless otherwise specified (see 6.2), tool holders shall be finished in accordance with manufacturers standard practice.

3.5 Interchangeability. All replacement parts shall be constructed to definite standards, tolerances, and clearances in order that any such parts of a particular type, class, or style may be replaced or adjusted without requiring modification.

3.6 Threads. All screw threads shall be in the inch system and conform to H28. Unified screw threads shall be used in lieu of American National screw threads and nonunified American Standard screw threads.

3.7 Identification marking. Each tool holder shall be marked in a permanently legible manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. In addition, offset holders shall be marked to indicate right or left-hand offset.

3.8 Type I, holder, lathe, turning (broached hole or bit channel parallel to base). Type I tool holders shall have the tool bit, broached hole or channel parallel to the base of the shank.

3.8.1 Class 1, straight shank (single size). Straight shank tool holders shall consist essentially of a body, tool-bit clamping device and a wrench. Class 1 holders shall be similar to figure 1 and shall conform to the dimensions of table I for type I, class 1 tool holders for number and size specified (see 6.2).

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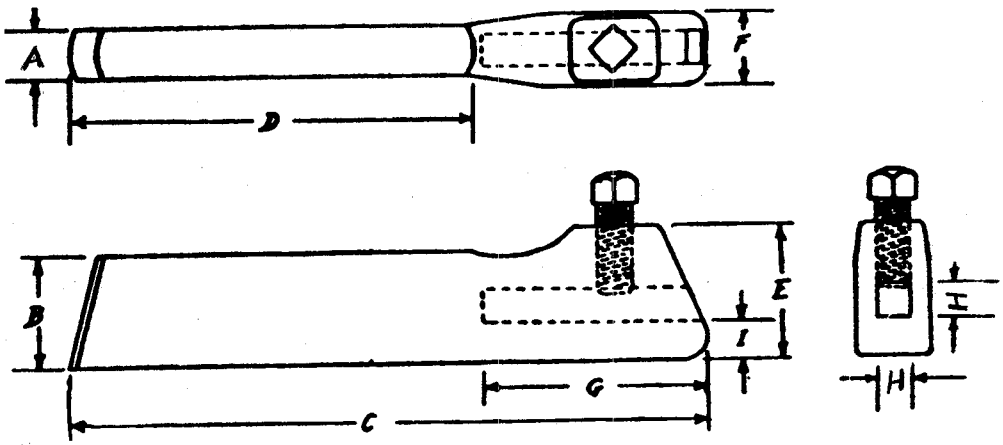


FIGURE 1. Type I, class 1, straight shank, style A, tool holder.

3.8.1.1 Body. The body shall have an integral shank and tool holding head, and shall be made of drop-forged alloy steel heat-treated to a Rockwell "C" 40 to 48 hardness or alloy steel, surface hardened to minimum 0.010 to 0.020 inch depth of full carburization as measured by microscope, and have a Rockwell "C" of 50 to 55 hardness.

3.8.1.1.1 Shank. The shank top and bottom surface shall be machine finished. The top surface shall be parallel to the bottom surface within 0.012 inch per foot. The bottom surface shall be flat and straight within 0.010 inch longitudinal concavity.

3.8.1.1.2 Head. The head shall be provided with a tool bit channel centrally aligned with longitudinal axis of shank, and parallel within a plus or minus 1° of base of shank. Means shall be provided for firmly securing tool bit in channel. The bottom of the tool bit channel length shall be flat within 0.002 inch. Eighty percent of the channel bottom surface shall be in contact with the tool bit so that no cracking or breaking of tool bits will result from pressure applied by clamping device. The front face of the head shall be chamfered at appropriate angles to provide maximum bit support, chip clearance, and visibility of cutting action.

3.8.1.2 Tool-bit clamping device. The clamping device shall be of alloy steel heat-treated to a Rockwell "C" 38 to 48 hardness or of alloy steel, surface hardened to a Rockwell "C" 40 to 45 hardness. Design shall insure securely clamped tool bit in channel by direct contact, or through medium of jaw or pressure pad. Clamping device shall not become detached from head nor become burred to prevent removal.

3.8.2 Class 2, left-hand offset shank.

3.8.2.1 Style A, rigid (single size). Left-hand offset shank, rigid style tool holders shall conform to 3.8.1, except that the axis of the head and tool-bit channel shall be rotated clockwise on a horizontal plane 34° ± 2°, from the shank. Class 2 holders shall be similar to figure 2 and shall conform to the dimensions of table I for type **L**, class 2 tool holders for number and size specified (see 6.2). The tool bit channel shall pass entirely through the head.

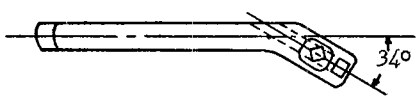


FIGURE 2. Types I and II, class 2, style A, left-hand offset shank, rigid style tool holders.

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3.8.2.2 Style B, multisize tool. Left-hand offset shank, multi-size tool holders shall conform to the requirements of 3.8.2.1, except that the tool holders shall be similar to figure 3. The dimensions shall conform to table II for class 2 tool holders for number and size specified (see 6.2).

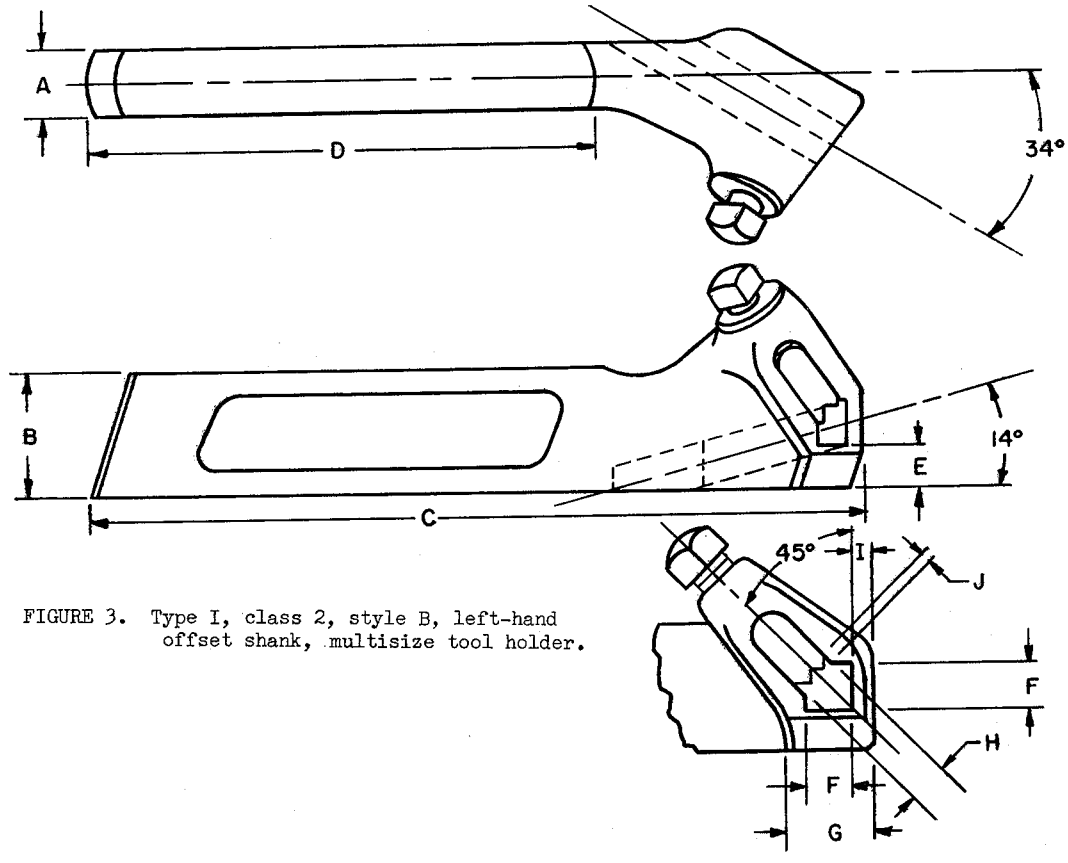


FIGURE 3. Type I, class 2, style B, left-hand offset shank, multisize tool holder.

3.8.3 Class 3, right-hand offset shank.

3.8.3.1 Style A, rigid (single size). Right-hand offset shank, rigid style holders shall conform to the requirements of 3.8.1, except that the axis of the head and tool-bit channel shall be rotated counter-clockwise on a horizontal plane 34° ± 2° from shank. Class 3 tool holders shall be similar to figure 4. The dimensions shall conform to table I for class 3 tool holders for number and size specified (see 6.2). Tool-bit channel shall pass entirely through head.

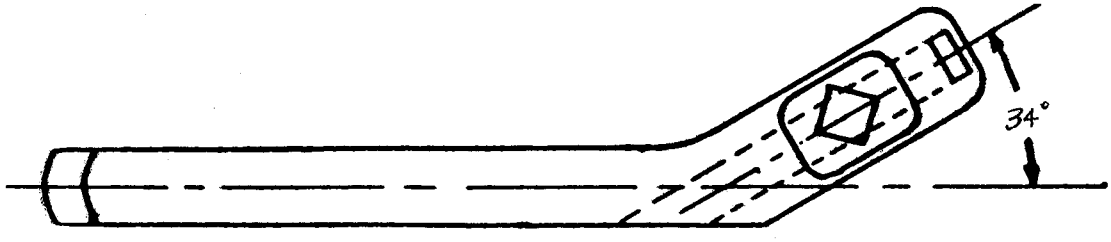


FIGURE 4. Types I and II, class 3, style A, right-hand offset shank, rigid style tool holder.

3.8.3.2 Style B, multisize tool. Right-hand offset shank, multi-size tool holders shall conform to the requirements of 3.8.3.1, except the tool holders shall be similar to figure 5. The dimensions shall conform to table II for class 3, style B tool holders for number and size specified (see 6.2 and 6.3).

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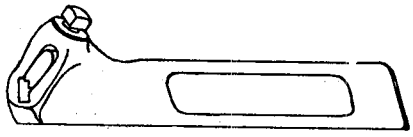


FIGURE 5. Type I, class 3, style B, right-hand offset shank, multisize tool holder.

3.9 Type II, holder, lathe, turning (14° tilted broached hole or bit channel). Type II tool holders shall conform to the requirements of 3.8.1, except the tool bit channel shall be inclined upward in a vertical plane 14° ± 1°, from base of shank.

3.9.1 Class 1, straight shank, rigid (single size). Straight shank, rigid style tool holders shall conform to the requirements of 3.8.1, except the tool bit channel shall be inclined upward in a vertical plane 14°, plus or minus 1°, from base of shank and shall be similar to figure 6. The dimensions shall conform to table III for type II, class 1 tool holders for number and size specified (see 6.2).

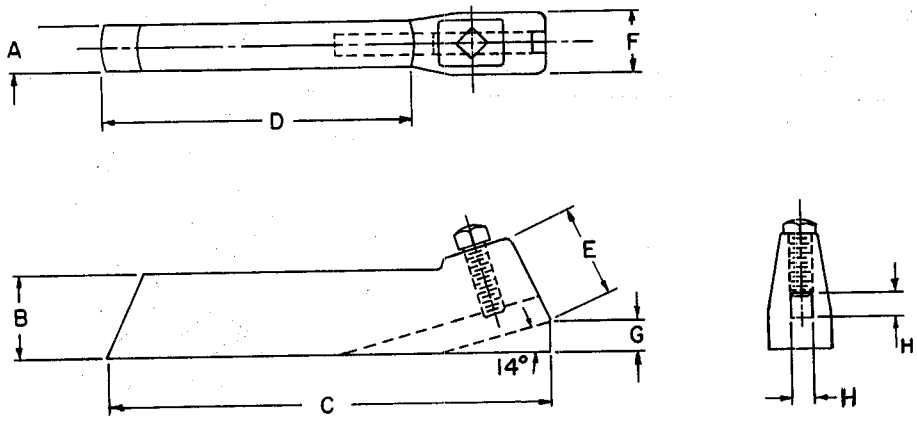


FIGURE 6. Type II, class 1, straight shank, rigid style tool holder.

3.9.2 Class 2, left-hand offset shank.

3.9.2.1 Style A, rigid (single size). Left-hand offset shank, rigid style tool holders shall conform to the requirements of 3.8.2.1, except that tool bit channel shall be inclined upward in a vertical plane 14° ± 1°, from base of shank and be similar to figure 2. The dimensions shall conform to table III for type II, class 2 tool holders for number and size specified (see 6.2).

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3.9.2.2 Style B, multisize tool. Left-hand offset shank, multi-size tool holders shall conform to the requirements of 3.8.2.2, except that tool bit channel shall be inclined upward in a vertical plane 14° ± 1°, from base of shank and shall be similar to figure 7. The dimensions shall conform to table IV for type II, class 2, style B tool holders for number and size specified (see 6.2).

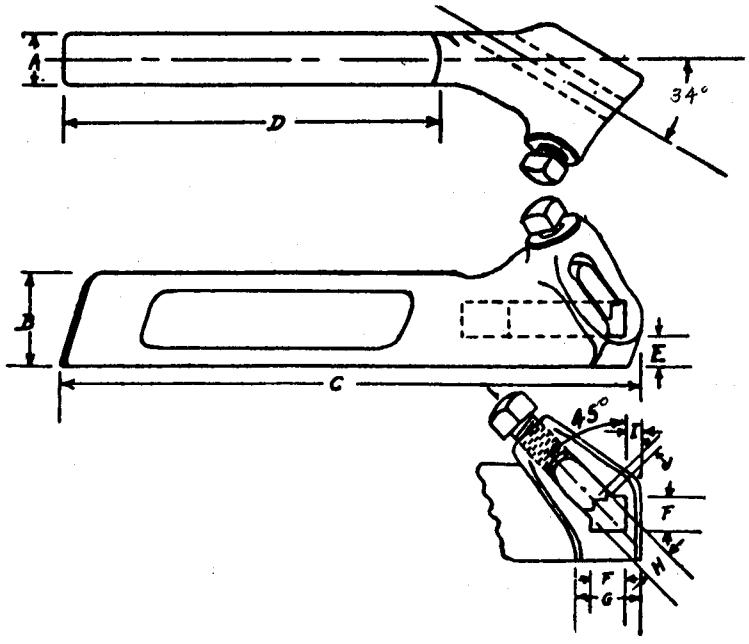


FIGURE 7. Type II, class 2, style B, left-hand offset shank, multisize tool style tool holders.

3.9.3 Class 3, right-hand offset shank.

3.9.3.1 Style A, rigid (single size). Right-hand offset shank, rigid style tool holders shall conform to the requirements of 3.8.3.1, except that tool bit channel shall be inclined upward in a vertical plane 14° ± 1°, from base of shank and shall be similar to figure 4. The dimensions shall conform to table IV for type II, class 3, style B tool holders for number and size specified (see 6.2).

3.9.3.2 Style B, multisize tool. Right-hand offset shank multi-size tool holders shall conform to the requirements of 3.8.3.2, except the tool holders shall be similar to figure 8. The dimensions shall conform to table IV for type II, class 3, style B tool holders for number and size specified (see 6.2).

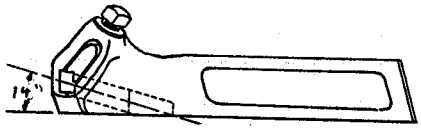


FIGURE 8. Type II, class 3, style B, right-hand offset shank, multisize tool holder.

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3.10 Type III, holder, cut-off or side cutting. Cut-off tool holders shall be of rigid style A or spring style B as specified (see 6.2), and shall consist essentially of a body, blade clamping device and wrench.

3.10.1 Body. The body shall have an integral shank and tool-holding head, and be of drop-forged alloy steel, heat-treated to Rockwell "C" 40 to 48 hardness, or alloy steel, surface hardened to minimum 0.010 to 0.020 inch depth of full carburization as measured by microscope, and have a Rockwell "C" 50 to 55 hardness.

3.10.1.1 Shank. The shank shall conform to the requirements of 3.8.1.1.1.

3.10.1.2 Head. The side of the head shall have a machined dovetail style groove to firmly support cut-off blade. The bottom, or dovetail surface of this groove shall be at an appropriate angle to hold cut-off blade against side of groove by wedging action, and shall be parallel with the bottom surface of the shank within 1-1/2°. The side of the groove shall be of the proper angle from vertical plane of head to provide side clearance for cut-off blade furnished in accordance with 00-C-851. The groove shall be of a width to accept the nominal height of cut-off blades for which intended, shall be true and straight, so that no cracking or breaking of blades may occur from clamping action. The front face of the head shall be chamfered at appropriate angles to provide maximum blade support, chip clearance and visibility of cutting action.

3.10.2 Cut-off blade clamping device. Cut-off blade clamping device shall be of steel, heat-treated to Rockwell "C" 40 to 60 hardness and finished to insure proper performance. The device shall securely clamp cut-off blade against side and bottom of groove in head, permitting no movement of blade during cutting action.

3.10.3 Class 1, straight shank.

3.10.3.1 Style A, rigid (single size). The blade holding groove of the straight shank, rigid style cut-off tool holders shall be in line with, and continuing through shank, on right-hand side, when viewed from forward, or cutting end, of tool holder. The tool holders shall conform to applicable dimensions of table V for number and size specified (see 6.2), and shall be similar to figure 9.

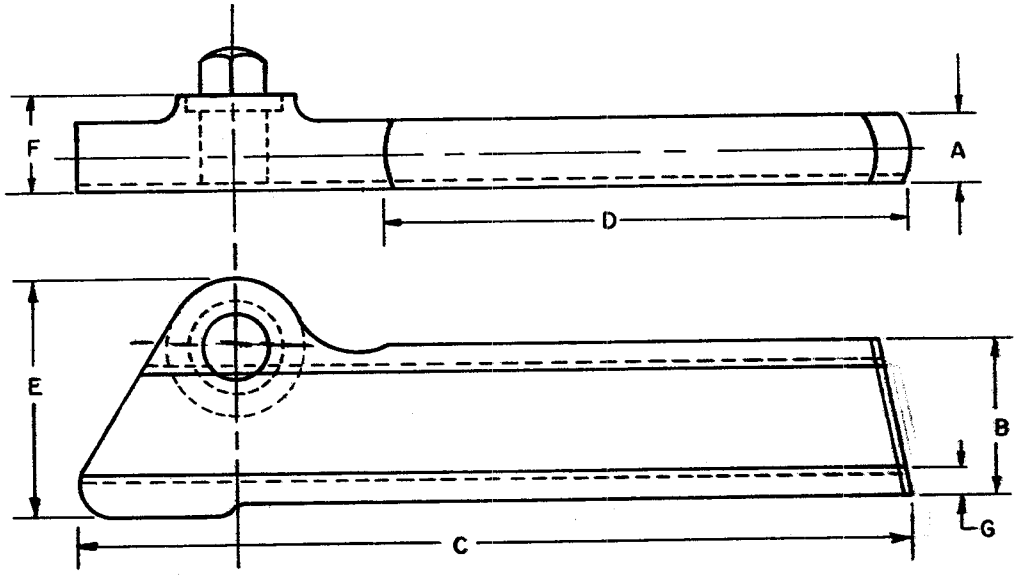


FIGURE 9. Type III, class 1, style A, straight shank, cut-off and side-cutting tool holder.

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3.10.3.2 Style B, spring. Style B, straight shank spring style cut-off tool holders shall conform to the requirements of 3.10.3.1, except that the head shall be a "gooseneck" form providing a spring action to cut-off blade. Clearance shall be provided at top and bottom of holding groove extending behind "gooseneck" and shall be similar to figure 10. The dimensions shall conform to table V for type III, class 1, style B tool holders for number and size specified (see 6.2).

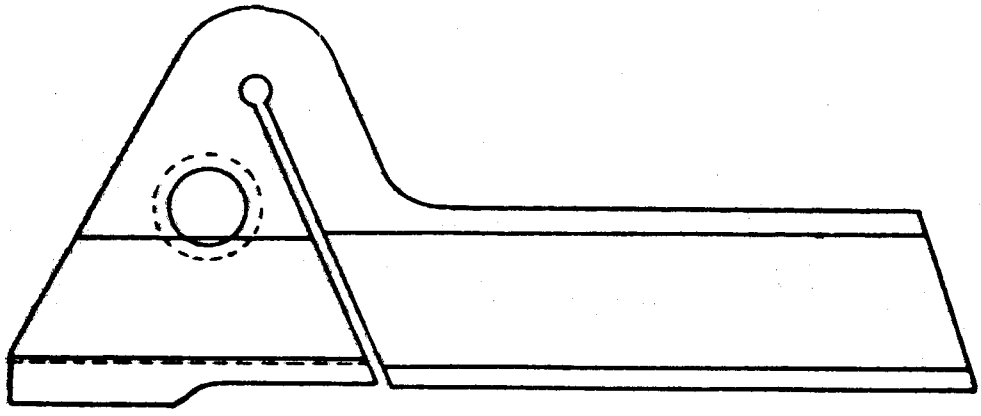


FIGURE 10. Type III, class 1, style B, spring style, straight shank, cut-off tool holder.

3.10.4 Class 2, left-hand offset shank.

3.10.4.1 Style A, rigid (single size). Left-hand offset shank, rigid style cut-off tool holders shall conform to the requirements of 3.10.3.1, except that the axis of the head and cut-off blade channel shall be turned counter-clockwise on a horizontal plane 34° ± 2°, from shank. The blade holding groove shall be on left-hand side when viewed from forward or cutting end of tool holder and shall be similar to figure 11. The dimensions shall conform to table V for type III, class 2, style A for number and size specified (see 6.2).

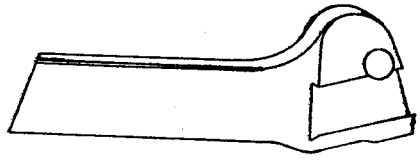


FIGURE 11. Type III, class 2, style A, left-hand offset shank, cut-off tool holders.

3.10.4.2 Style B, spring. Style B, left-hand offset shank, spring style, cut-off tool holders shall conform to the requirements of 3.10.4.1, except that the head shall be of "gooseneck" form providing a spring action to cut-off blade. Clearance shall be provided at top and bottom of holding groove extending behind "gooseneck" and shall be similar to figure 12. The dimensions shall conform to table V for type III, class 2, style B for number and size specified (see 6.2).

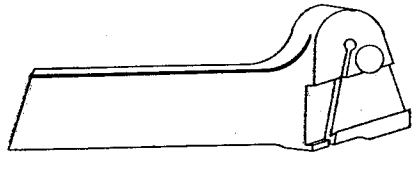


FIGURE 12. Type III, class 2, style B, spring style, left-hand offset shank, cut-off tool holder.

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3.10.5 Class 3, right-hand offset shank.

3.10.5.1 Style A, rigid (single size). Style A, right-hand offset shank, rigid style, cut-off tool holders shall conform to the requirements of 3.10.3.1, except that the axis of the head and tool bit channel shall be bent clockwise, on a horizontal plane 34° ± 2°, from the shank and shall be similar to figure 13. The dimensions shall conform to table V for type III, class 3, style A for number and size specified (see 6.2).

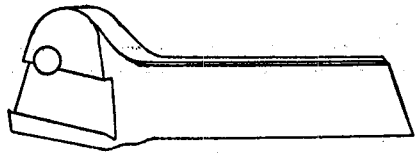


FIGURE 13. Type III, class 3, style A, right-hand offset shank, cut-off tool holders.

3.10.5.2 Style B, spring. Style B, right-hand offset shank, spring style, cut-off tool holders shall conform to the requirements of 3.10.5.1, except that the head shall be of a "gooseneck" form providing a spring action to cut-off blades and shall be similar to figure 14. Clearance shall be provided at top and bottom of holding groove extending behind "gooseneck" and shall be similar to figure 14. The dimensions shall conform to table V for type III, class 3, style B for number and size specified (see 6.2).

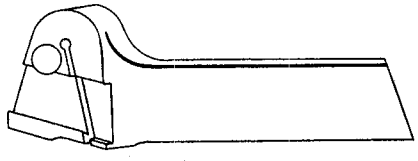


FIGURE 14. Type III, class 3, style B, spring style, right-hand offset shank, cut-off tool holder.

3.11 Type IV, holder, lathe, threading. Threading tool holders shall consist essentially of a body, tool clamping device, threading tool and a wrench.

3.11.1 Body. The body shall have an integral shank and tool holding head, and be of a drop-forged alloy steel, heat-treated to a Rockwell "C" 40 to 48 hardness or alloy steel, surface hardened to minimum of 0.010 to 0.020 inch depth of full carburization as measured by microscope, and have a Rockwell "C" 50 to 55 hardness.

3.11.1.1 Shank. The shank shall conform to the requirements of 3.8.1.1.1.

3.11.1.2 Head. The head shall be so designed as to firmly support the tool by the clamping device.

3.11.2 Clamping device. The clamping device shall be such as to firmly support tool in or on head, permitting no movement of tool duration cutting action, nor cracking or breaking of tool when clamped with clamping device.

3.11.3 Class 1, rigid. Rigid threading tool holders shall conform to the requirements of 3.11, except the head shall be bent clockwise on a horizontal plane 34°, plus or minus 2°, from shank. Face of head upon which tool is clamped shall be perpendicular within 2° from base of shank. A horizontally tapped boss fitted with adjusting screw to control tool edge height shall be provided on the side of the head and shall be similar to figure 15. The dimensions shall conform to table VI for type IV, class 1 holders for number and size specified (see 6.2).

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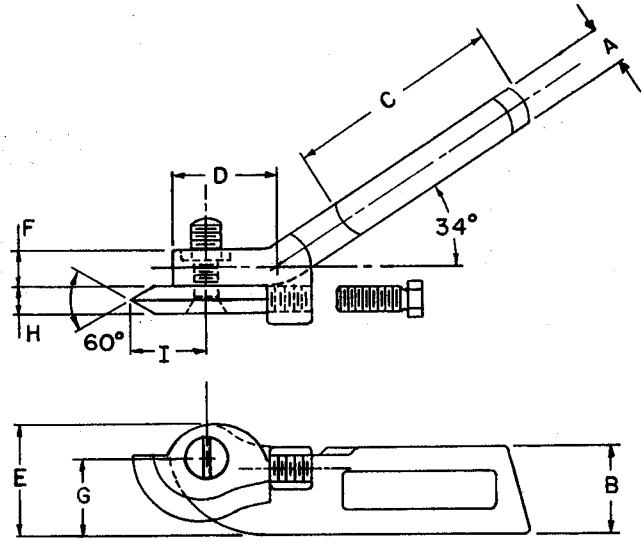


FIGURE 15. Type IV, class 1, rigid threading tool holder.

3.11.3.1 Threading tools. Threading tools for use with class 1 tool holder shall be single point, radial form, of high-speed tool steel hardened to a Rockwell "C" 63 to 67 hardness. All threading tools shall be parallel in thickness to within plus or minus 0.005 inch and shall be ground to an included angle of 60°, +0°, -1°, on the periphery, and this angle shall be properly relieved to provide clearance below cutting edge. Threading tools shall conform to applicable requirements of table VI. Unless otherwise specified (see 6.2 and 6.9), a 60° included angle, sharp V-thread radial form threading tool shall be furnished with each tool holder.

3.11.4 Class 2, spring style. Holders of class 2 spring style tool holders shall have a "gooseneck" form of head to provide a resilient tool mounting, but means shall be provided to lock or clamp tool in a rigid position when such is required. Holders shall conform to applicable dimensions of table VII, and be similar to figure 16.

3.11.4.1 Shank. Shank shall conform to the requirements of 3.8.1.1.1.

3.11.4.2 Tool supporting head. The head shall be machined so as to securely support the cutter bit parallel, within plus or minus 2° to the base of the shank and in various angular positions around the axis of the head.

3.11.4.3 Tool-bit clamping device. The clamping device shall be of alloy steel heat-treated to a Rockwell "C" 38 to 48 hardness or of alloy steel, surface hardened to a Rockwell "C" 40 to 45 hardness. Design shall insure securely clamped tool bit in channel by direct contact or through medium of jaw or pressure pad. Clamping device shall not become detached from head nor become burred to prevent removal.

3.11.4.4 Threading tools. Threading tool for use with class 2 tool holder square single point tool bit shall conform to the applicable requirements of table VII, of high-speed steel hardened to a Rockwell "C" 63 to 67 hardness. Unless otherwise specified (see 6.2), a 60° included angle sharp V-thread single point threading tool bit shall be furnished with each tool holder.

3.11.4.5 Wrenches. A wrench to fit the clamping device shall be furnished with each holder of this style.

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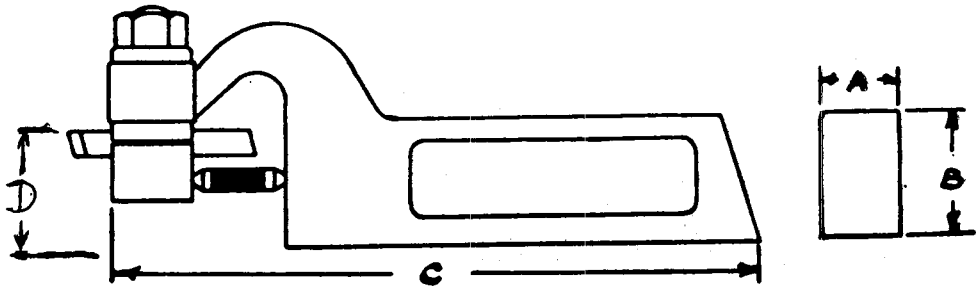


FIGURE 16. Type IV, class 2, spring-style threading tool holder.

3.12 Type V, holder, lathe, knurling. Knurling tool holder shall consist essentially of a body, knurl holder, support pins, and knurls.

3.12.1 Body. The body shall have an integral shank and knurl holder head, and be of drop-forged steel, heat-treated to a Rockwell "C" 40 to 48 hardness, or case-hardened steel, surface hardened 0.010 to 0.012 inch deep to a Rockwell "C" 50 to 55 hardness.

3.12.1.1 Shank. The shank shall conform to the requirements for type I, class 1 shanks (see 3.8.1.1.1).

3.12.1.2 Head. The head shall be designed so as to support a swiveling, self-centering knurl holder with adequate strength to resist axial and lateral thrusts.

3.12.2 Knurl holder. The knurl holder shall be of steel, heat-treated and finished to insure proper performance and resistance to wearing action of knurls. Means shall be provided for the mounting and replacing of the knurls. Design of holder shall permit use of any work diameter, and thickness of supports of knurls shall be of the optimum dimension compatible with the strength requirements.

3.12.3 Support pins. Support pins shall be of steel, heat-treated and finished to properly support and permit free rotation of knurls or rotating heads. Pins shall be finished to such limits so as to be interchangeable with pins of the same size and type of knurling tool holder of any one manufacturer.

3.12.4 Knurls. Knurls for bench and engine lathe tool holders shall conform to USASI B94.6. Knurls shall be of tool steel and hardened to a Rockwell "C" 63 to 67 and finished to insure proper performance. The teeth shall be sharp and concentric and parallel to the bore and shall conform to the applicable dimensions of table VIII. Knurls shall be similar to figure 17, shall be furnished in pairs, in either a diamond or straight line pattern, coarse, medium, or fine, 45° bevel or full face as specified (see 6.2 and 6.7).

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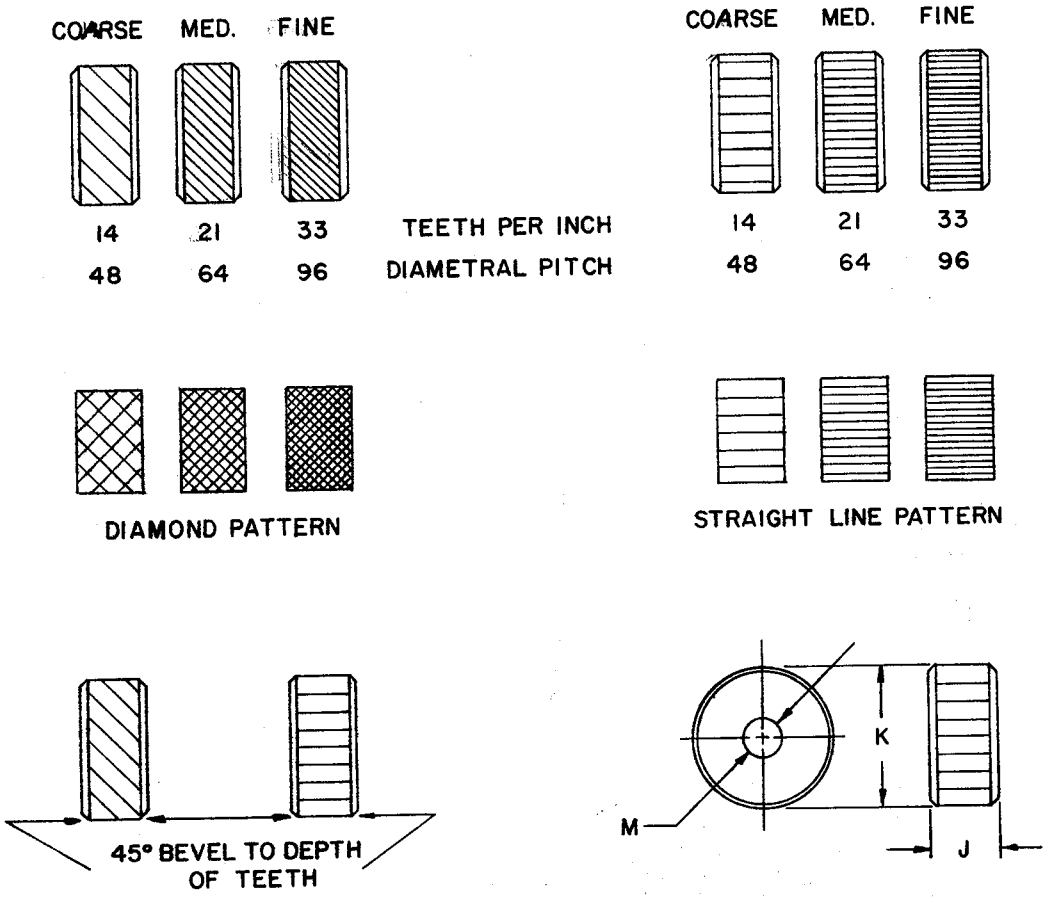


FIGURE 17. Knurls for type V, classes 1 and 2, knurling tool holders.

3.12.5 Class 1, single. Single-knurling tool holders shall be of a single, swivel-head style having one pair of replaceable knurls. Holders shall conform to the applicable dimensions of table IX, and be similar to figure 18.

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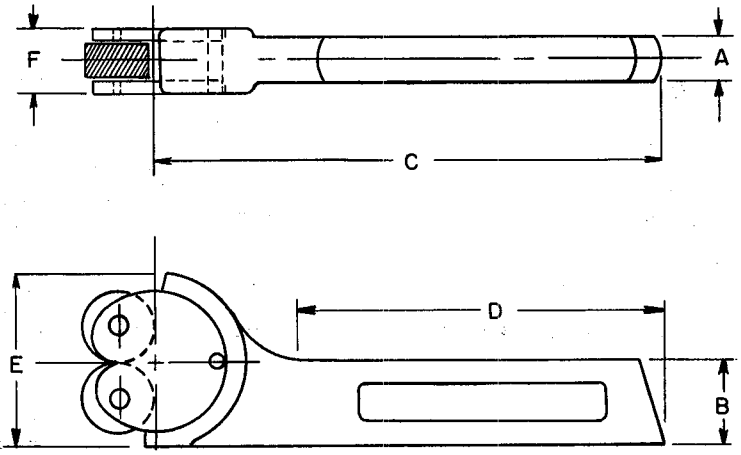


FIGURE 18. Type V, class 1, single-knurling tool holder.

3.12.5.1 Knurls. Knurls shall conform to USASI B94.6 and applicable dimensions of table VIII. Unless otherwise specified (see 6.2 and 6.7), each knurling tool holder of type V, class 1, shall be furnished with a pair of medium diamond pattern knurls.

3.12.6 Class 2, multiple. Multiple-knurling tool holders shall have a single, rotating head containing three pairs of replaceable knurls. Holders shall conform to applicable dimensions of table X, and be similar to figure 19.

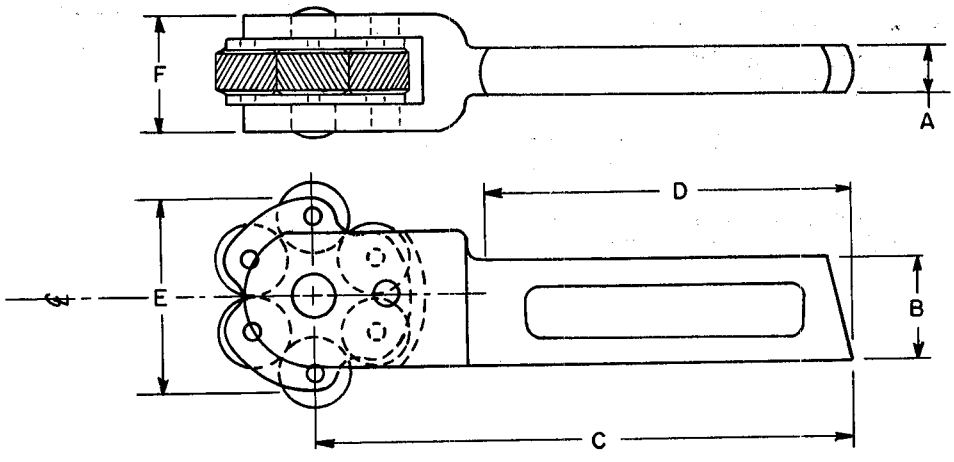


FIGURE 19. Type V, class 2, multiple-knurling tool holder.

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3.12.6.1 Knurls. Knurls shall conform to USASI B94.6. Unless otherwise specified (see 6.2 and 6.7), each knurling tool holder of type V, class 2 shall be furnished with one pair each of coarse, medium, and fine pitch diamond pattern knurls.

3.13 Type VI, holder, lathe, boring. Boring tool holders shall provide means of firmly securing boring tools or bars to the tool post holder parallel with the center line of the lathe. Holders shall provide maximum strength and maximum visibility of work at a minimum of interference with the boring operation, while precluding any movement of bar or tool in the clamping device druing operation.

3.13.1 Boring tools. Solid forged boring tools shall be of high-speed steel, properly forged to shape and hardened for approximately 1/8 inch behind cutting edge to a Rockwell "C" 63 to 67 hardness. The tool shall be ground, ready for use, shall conform to applicable dimensions of table XI and be similar to figure 20.

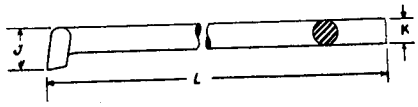


FIGURE 20. Solid-forged boring tools.

3.13.2 Boring bars. Boring bars shall be of steel, machined and finished to insure proper performance. Bars shall provide means for securely holding tool bits as near the end of bar as practical, and at the angles and locations given for the respective styles. The bars shall conform to applicable dimensions of table XII and be similar to figure 21.

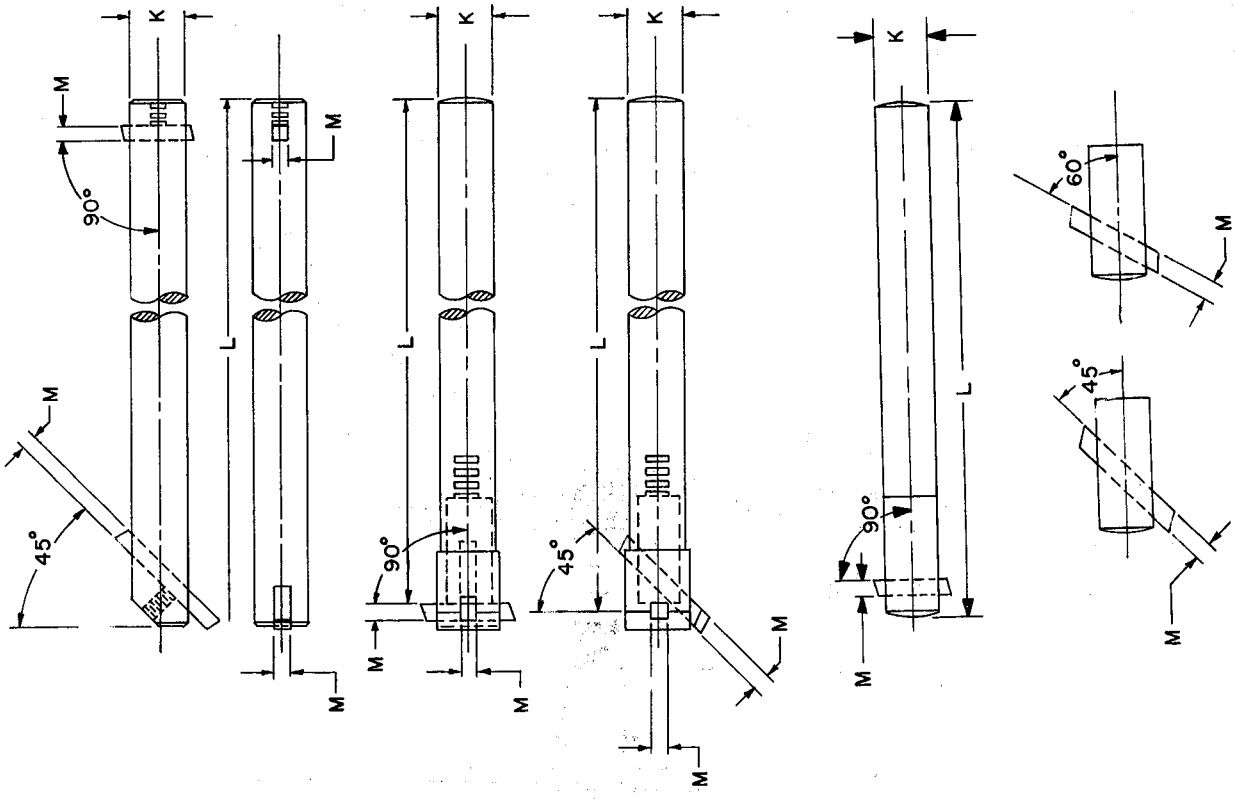


FIGURE 21. Plain, sleeve and end-cap boring bars with square tool bits.

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3.13.2.1 Plain bars. Plain style boring bars shall have a tool-bit channel at 90° and another at 45°, both plus or minus 2°, from the axis of the bar and at opposite ends of the bar.

3.13.2.2 Sleeve bars. Sleeve style boring bars shall have two tool-bit channels in the bar sleeve as specified for plain bars except that the channels shall be 90° apart, both at the same end of bar sleeve. Tool bits shall be securely clamped in sleeve by threaded stud.

3.13.2.3 End-cap bars. End-cap-style boring bars shall have three interchangeable end caps with tool bit channels at 90°, 45° and 60°, respectively, all plus or minus 2° from the axis of the bar. Tool bits shall be securely fastened in end caps by threaded end of boring bar.

3.13.2.4 Wrenches. Unless otherwise specified (see 6.2), wrenches shall be as specified in 3.3.

3.13.3 Class 1, shank mounted. Shank mounted, boring tool holders shall consist essentially of a body, clamping device and boring tools or bars and wrenches.

3.13.3.1 Body. The body shall have an integral shank and tool holding head, and be of drop-forged alloy steel, heat-treated to a Rockwell "C" 40 to 52 hardness, or alloy steel, surface hardened to minimum 0.010 to 0.020 inch depth of full carburization as measured by microscope, and have a Rockwell "C" 50 to 55. (The style "C", split clamp tool holding head shall have a hardness of "C" 26 to 34.)

3.13.3.1.1 Head. The tool holding head shall be so designed as to properly accommodate the capacity range of boring bars for the respective size of tool holder.

3.13.3.1.2 Shank. The shank shall conform to 3.8.1.1.1.

3.13.3.2 Clamping device. The clamping device shall be of drop-forged alloy steel integral with head or attached thereto. Device shall be so designed as to rigidly and firmly clamp boring bars or tools to holding head and may clamp directly on bar or through medium of gib or bushing.

3.13.3.3 Clamp screws. The clamping screws shall be of alloy steel, heat-treated to a Rockwell "C" 38 to 45 hardness, and shall be finished to a suitable size and fit to insure proper performance. Clamp screws or nuts shall be replaceable and shall securely clamp boring tools or bars in body or support, and shall not become detached from holder.

3.13.3.4 Style A, reversible. Reversible, shank-mounted, boring and turning tool holders shall have a tool holding head grooved on top and bottom for supporting boring tools and tool bits with a yoke style clamping device. The holders shall conform to applicable dimensions of table XIII and be similar to figure 22 (see 6.2 and 6.4).

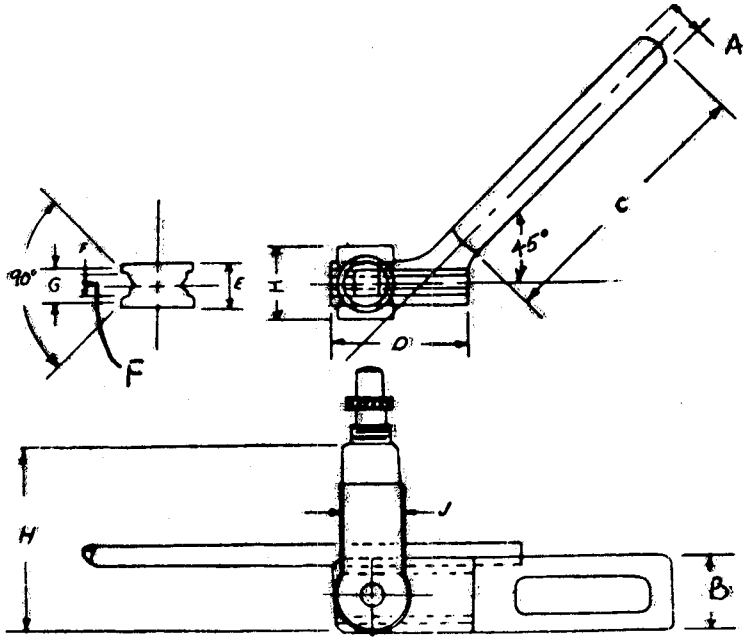


FIGURE 22. Type VI, class 1, style A reversible, shank-mounted boring and turning tool holder.

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3.13.3.4.1 Holder, boring tool. Each type VI, class 1, style A tool holder shall be furnished with two boring tools conforming to applicable dimensions of table XI and appropriate square turning tool bit.

3.13.3.5 Style B, clamp block. Clamp block, shank-mounted, boring tool holders shall have a reversible clamp block to fasten boring bars to tool-holding head. The holders shall conform to applicable dimensions of table XIV, and be similar to figure 23.

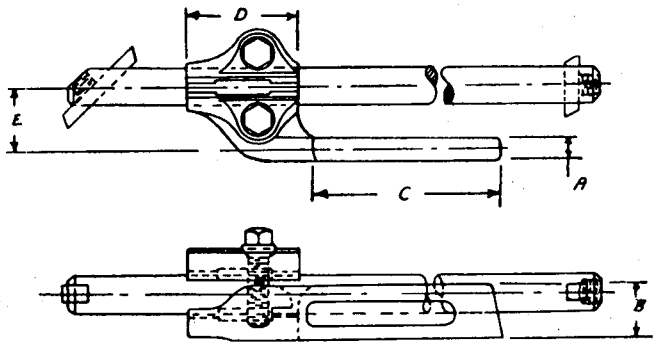


FIGURE 23. Type VI, class 1, style B, clamp block, shank-mounted boring tool holders.

3.13.3.5.1 Boring bars. Each tool holder shall be furnished with one sleeve type boring bar of the maximum size within the capacity of the holder conforming to applicable dimension of table XI for the holder number specified (see 6.2).

3.13.3.5.2 Wrenches. A wrench to fit the clamping device and a wrench to fasten tool bits to boring bars shall be furnished with each tool holder of this style.

3.13.3.6 Style C, split clamp. The tool-holding head of the split clamp, shank-mounted boring tool holders shall be hardened to a Rockwell "C" 26 to 34. The holders shall conform to applicable dimensions of table XV for the tool holder number (see 6.2), and shall be similar to figure 24.

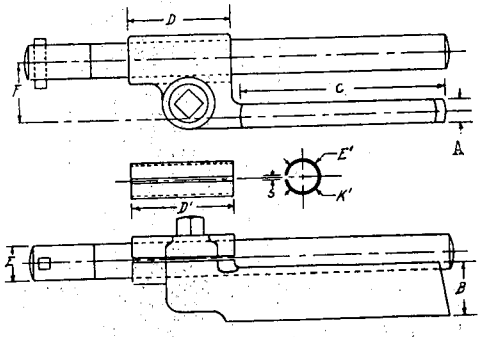


FIGURE 24. Type VI, class 1, style C, split clamp, shank-mounted boring tool holder and bushing.

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3.13.3.6.1 Boring bars. Each tool holder shall be furnished with one boring bar and appropriate size tool bit of the maximum size within the capacity of the tool holder unless a definite size is furnished with an end cap bar (see 6.5). Two appropriate size tool bits shall be furnished with each plain bar and three appropriate size tool bits with each end cap bar conforming to applicable dimensions of table XII.

3.13.3.6.2 Wrenches. A combination wrench to fit the clamping device and to fasten tool bits in boring bars (see 3.3), shall be furnished with each tool holder of this style, except that a separate wrench for each purpose shall be furnished if plain style boring bars are specified (see 6.2).

3.13.3.6.3 Bushings. Bushings for use with various sizes of boring bars in this style of tool holders shall be of split section, of steel and finished to insure proper performance. Bushings shall conform to applicable dimensions of table XV and be similar to figure 24 (see 6.6).

3.13.4 Class 2, T-slot mounted. T-slot mounted boring bar holders shall consist essentially of a T-slot mounting, bar support, clamping device, boring bars, tool bits, and wrenches.

3.13.4.1 T-slot mounting. Mounting shall be of steel, finished so as to securely attach boring bar holder to tool post T-slot of lathe.

3.13.4.2 Bar support. Support shall be of steel and constructed so as to support various size boring bars parallel to center line of lathe.

3.13.4.3 Clamping device. The clamping device shall be of steel and constructed so as to securely clamp boring bars to bar support with no distortion of bar.

3.13.4.4 Style A, adjustable height. The bar support of the adjustable height, T-slot mounted, boring bar holder shall provide means of adjusting height of boring bar. Holders shall conform to applicable dimensions of table XVI, and be similar to figure 25.

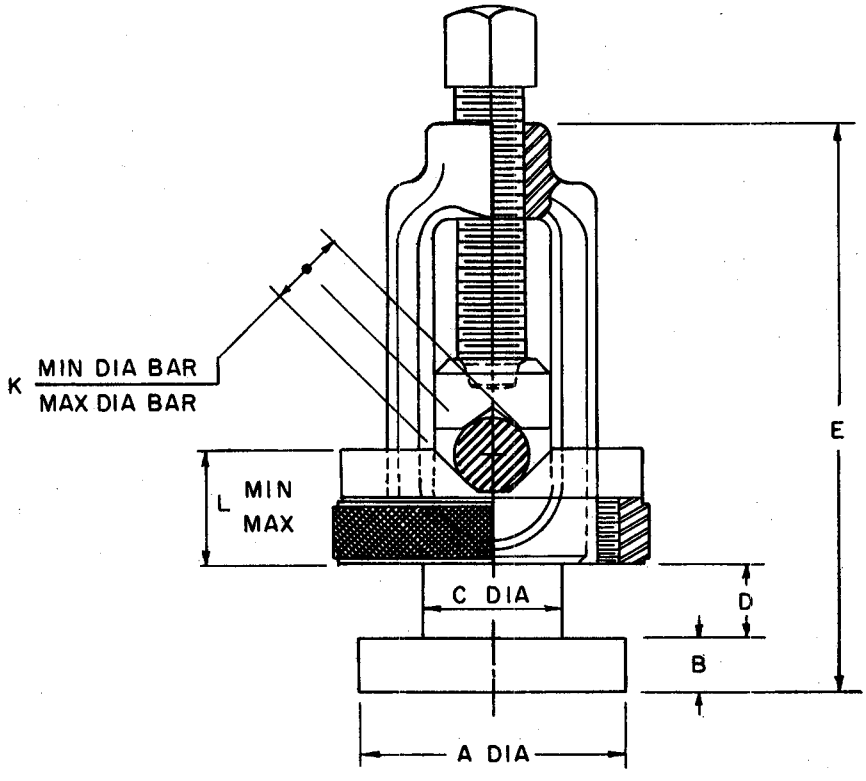


FIGURE 25. Type VI, class 2, style A, adjustable height, boring tool holder.

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3.13.4.4.1 Boring bars. Each tool holder shall be furnished with one boring bar of the maximum size within the capacity of the holder conforming to the applicable dimensions of table XVI. If no definite size is specified, the holder shall be furnished with a sleeve type bar (see 6.5).

3.13.4.4.2 Wrenches. A wrench to fit the clamping device and one to fasten tool bits to boring bars (see 3.3), shall be furnished with each tool holder of this style.

3.13.4.5 Style B, fixed height. Fixed height, T-slot mounted, boring bar holders shall conform to applicable dimensions of table XVII, and be similar to figure 26.

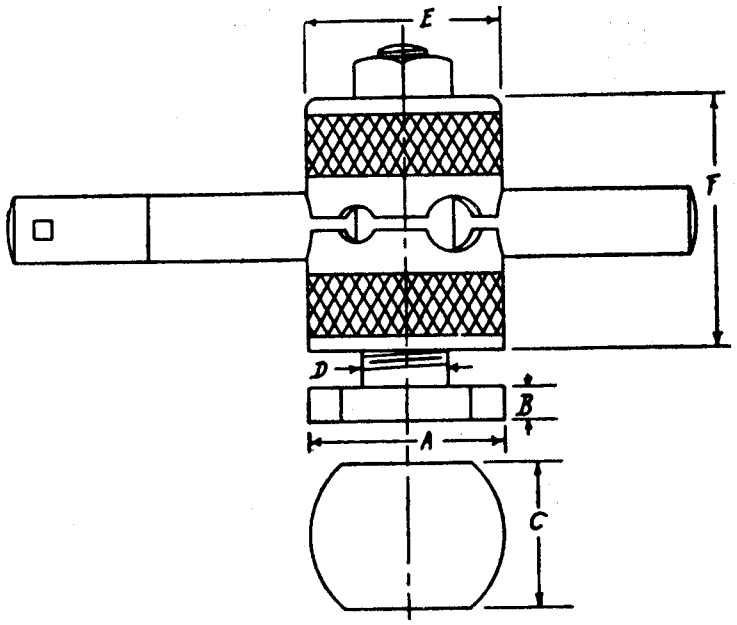


FIGURE 26. Type VI, class 2, style B, fixed height, boring tool holders.

3.13.4.5.1 Boring bars. Each holder shall be furnished with three boring bars conforming to the applicable dimensions of table XVII. Each bar shall be equipped with its full complement of tool-bit holding heads. A solid boring tool shall be furnished for the 3/8 inch diameter bar conforming to the applicable dimensions of table XI for holder number G-L-50.

3.13.4.5.2 Wrenches. A wrench to fit the clamping device and one to fasten tool bits to boring bars, or a combination (see 3.3) wrench shall be furnished with each holder of this style.

3.14 Type VII, holder, planer and shaper, adjustable. Adjustable planer and shaper tool holders shall consist essentially of a body, clamping device, tool bit and wrench, shall conform to applicable dimensions of table XVIII, and be similar to figure 27.

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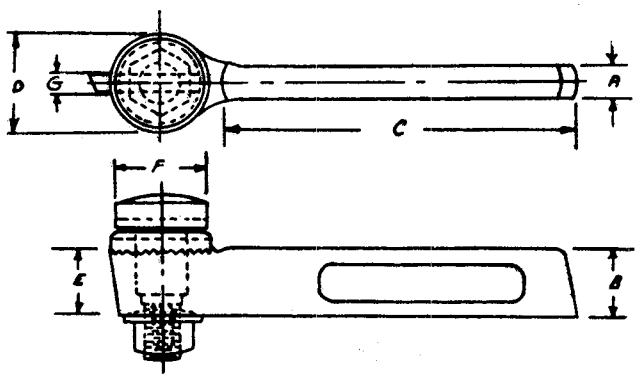


FIGURE 27. Type VII, adjustable planer and shaper, tool holder.

3.14.1 Body. The body shall have an integral shank and tool supporting head, shall be of drop-forged alloy steel, heat-treated to a Rockwell "C" 40 to 48 hardness or alloy steel surface hardened to minimum 0.010 to 0.020 inch depth of full carburization as measured by microscope and have a Rockwell "C" 50 to 55 hardness.

3.14.1.1 Shank. The shank shall conform to 3.8.1.1.1.

3.14.1.2 Tool supporting head. The head shall be machined so as to securely support cutter bits parallel, within plus or minus 2°, to the base of the shank, and in various angular positions around the axis of the head.

3.14.2 Clamping device. The clamping device shall be of steel and constructed so as to securely and rigidly clamp tool bit to supporting head at any predetermined position, with no tendency to crack or break tool bits, providing maximum strength and visibility of work, and minimum interference with cutting operation.

3.14.3 Wrenches. A wrench to fit the clamping device shall be furnished with each tool holder of this type.

3.15 Type VIII, holder, shaper, extension. Extension-bar shaper tool holders shall consist essentially of a body, extension bar, clamping device, and when specified (see 6.2 and 6.8), bushings. Holders shall conform to applicable dimensions of table XIX, and be similar to figure 28.

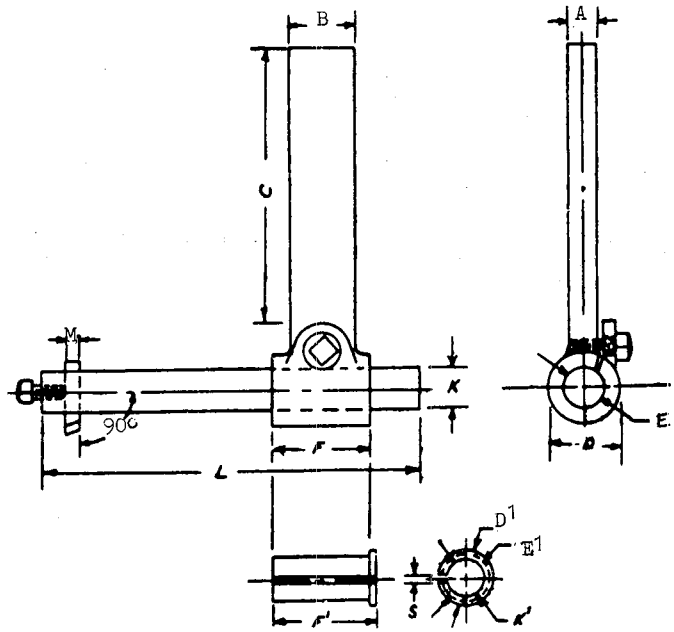


FIGURE 28. Type VIII, extension-bar shaper, tool holder and bushing.

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3.15.1 Body. The body shall have an integral shank and bar holding head, shall be of drop-forged alloy steel heat-treated to a Rockwell "C" 40 to 48 hardness on the shank and "C" 26 to 34 hardness on the head, or alloy steel, surface hardened to minimum 0.010 to 0020 inch depth of full carburization as measured by microscope, and have a Rockwell "C" 50 to 55 hardness on the shank and have a Rockwell "C" 40 to 45 hardness on the head.

3.15.1.1 Shank. The shank shall conform to 3.8.1.1.1.

3.15.1.2 Bar holding head. The head shall be split spring section style, provided with a machined bore perpendicular within a plus or minus 1°, to the base of the shank, providing an adjustable support for the extension bar.

3.15.2 Extension bar. Extension bars shall be of steel, conforming to the applicable dimensions of table XX, and provided with a tool-bit channel of 90° ± 2°, to the axis of the bar and as near the end of the bar as practicable. Each tool holder shall be furnished with one extension bar of the maximum size within the capacity of the holder unless a definite size is specified (see 6.2 and 6.8).

3.15.3 Clamping device. Means shall be provided for clamping the extension bar in the head and the tool bit in the channel of the extension bar.

3.15.4 Bushings. Bushings for use with various sizes of extension bars in the tool holders shall be of split section, of steel and finished to insure alignment of bar, within a plus or minus 1°. Bushings shall conform to applicable dimensions of table XIX, and be similar to figure 28.

3.15.5 Wrenches. One wrench to fit clamping device of bar holding head and one to fit clamping device of tool-bit channel shall be furnished with each tool holder of this type.

3.16 Workmanship. The workmanship of tool holders and boring bars and accessories shall be of the quality prevailing among manufacturers normally producing tool holders and boring bars as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All tool holders of the same size, type, class, and style presented at one time for delivery shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sample for quality conformance. A random sample of tool holders shall be selected from each lot in conformance with MIL-STD-105 at Inspection Level S-3.

4.3 Quality conformance inspection. Quality conformance inspection shall be performed on each sample tool holder selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

(a) Examination (see 4.4).
(b) Tests (see 4.5).
(c) Examination of preservation, packaging, packing, and marking (see 4.6).

4.4 Examination. Each tool holder selected in accordance with 4.2.2 shall be examined to determine compliance with this specification. Examination shall be conducted in accordance with table XXI. The acceptable quality level (AQL) shall be as indicated in table XXI. Nonconforming tool holders shall not be offered for delivery and if the nonconforming tool holders in any one sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

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4.5 Hardness. Each of the tool holders selected in accordance with 4.2.2 shall be tested for Rockwell hardness to verify compliance with this specification. If any sample tool holder does not meet the requirements specified herein it shall not be offered for delivery and if the number of nonconforming tool holders in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

4.6 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table XXII. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105, inspection level S-2, with an AQL of 4.0 defects per hundred units. Defects of closure shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each holder and component parts including wrench, shall be packaged together in accordance with MIL-P-116, method 1C.

5.1.2 Level B. Each holder and component parts including wrench shall be packaged in a snug-fitting paperboard box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.3 Level C. Packaging shall be in a manner which will afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Holders, packaged as specified in 5.1 shall be packed in a snug-fitting wood-cleated plywood or fiberboard box conforming to PPP-B-601, overseas type, or PPP-B-636, class weather-resistant.

5.2.2 Level B. Holders, packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class domestic. Gross weight of the shipping container shall not exceed 65 pounds.

5.2.3 Level C. Holders, packaged as specified in 5.1 shall be packed in a container which will afford adequate protection against damage during shipment from the supply source to the first receiving activity. Supplier may use his standard practice when it meets this requirement.

5.3 Marking. Marking shall be as specified in 5.3.1 and 5.3.2 (see 6.2).

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123. Additionally, each unit and intermediate package shall be marked with Federal Stock Number, Nomenclature and quantity.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Tool holders are intended for use on engine lathes and also are adaptable for use on shapers and planers in certain applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2.1).
(c) Finish, if different (see 3.4).
(d) Tool holder number required (see tables I, III and V through XX).
(e) Tool holder number and size required (see tables II and IV).
(f) Threading tool furnished with each tool holder, if different (see 3.11.3.1).
(g) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(h) Marking required (see 5.3).

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6.3 Boring tools and bars. When boring tools or bars are desired for use with type I, class 3, style B tool holders, the style number of the bar, if any, or the type, diameter and length of the boring tool or bar and the tool holder number for which it is intended, should be specified in the contract or order (see 6.2).

6.4 Boring tools. When additional solid-forged boring tools are desired for use with type VI, class 1, style A boring tool holders, the diameter and length of the boring tool and the tool holder number for which it is intended should be specified in the contract or order (see 6.2).

6.5 Boring bars. When additional boring bars for use with type VI, class 1, style B or C; or type VI, class 2, tool holders are desired with or without holders, the number, type, diameter and length of bar, and the tool holder number for which it is intended should be specified in the contract or order (see 6.2).

6.6 Bushings. When bushings are desired for use with type VI, class 1, style C, or type VIII tool holders, the number and size of the bushing, and the tool holder number for which it is intended should be specified in the contract or order (see 6.2).

6.7 Knurls. When type V, class 1 or 2 knurling tool holders are desired with other than the knurls specified in 3.10.6.1 and 3.10.7.1, the pitch, pattern, face, and tool holder number for which it is intended should be specified in the contract or order (see 6.2).

6.8 Extension bars. When additional extension bars and bushings for use with type VIII tool holders are desired with or without holders, the number of the bar and the tool holder for which it is intended, should be specified in the contract or order (see 6.2). Each bar shall be furnished with a wrench in accordance with 3.3.

6.9 Threading tools. When additional threading tools for use with type IV, class 1, tool holders are desired with or without holders, the form, pitch, and the tool holder number for which it is intended should be specified in the contract or order (see 6.2).

6.10 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of tool holders and accessories otherwise conforming to this specification.

6.11 Changes from previous issue. The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes from the previous issue of this documents.

MILITARY INTEREST CUSTODIANS:

Army - WC
Navy - SH
Air Force - 84

Review activities:

Army - WC
Navy - SH, YD
Air Force - 84

User activity:

Navy - MC

Preparing activity:

Navy - SH

CIVIL AGENCIES INTEREST:

GSA-FSS

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TABLE I. Type I, class 1, and style A of classes 2 and 3, tool holders

Tool holder number			Size of square tool bit	Thickness of shank			Depth of shank			Overall length of holder			Length of shank	Overall height of holder		Width of holder head		Length of square broached hole type I class 1 only	Size of broached square hole		Height to broached square hole	
Type I				dimension A			dimension B			dimension C			dimension D	dimension E		dimension F		dimension G	dimension H		dimension I	
Class 1	Class 2 Style A	Class 2 Style A	Inch	Inches			Inches			Inches			Inches	Inches		Inches		Inches	Inch		Inch	
			Nom.	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Min.	Max.	Min.	Max.
A-S-10	A-L-20	A-R-30	1/4	3/8	0.35	0.39	15/16	0.90	0.96	6	5	6-1/2	3-11/16	1-1/16	1-3/16	17/32	39/64	1-7/8	0.260	0.275	7/16	1/2
A-S-11	A-L-21	A-R-31	5/16	1/2	.48	.53	1-1/4	1.21	1.28	7	6	7-1/2	4-3/4	1-3/8	1-1/2	21/32	23/32	2-3/8	.327	.338	17/32	5/8
A-S-12	A-L-22	A-R-32	3/8	5/8	.60	.66	1-1/2	1.40	1.60	8	7	8-1/2	5-1/4	1-9/16	1-13/16	25/32	55/64	2-3/4	.390	.400	19/32	3/4
A-S-13	A-L-23	A-R-33	7/16	3/4	.73	.79	1-3/4	1.65	1.82	9	8	9-1/2	5-3/4	1-15/16	2-1/16	29/32	63/64	3-1/8	.453	.467	11/16	13/16
A-S-14	A-L-24	A-R-34	1/2	7/8	.85	.92	1-7/8	1.75	2.00	10	9	10-1/2	6-1/2	2-3/16	2-5/16	1-1/32	1-7/64	3-3/4	.515	.530	23/32	7/8
A-S-15	A-L-25	A-R-35	5/8	1	.97	1.05	2-1/8	2.00	2.25	12	11	12-1/2	7-3/4	2-9/16	2-11/16	1-11/32	1-15/32	4-1/4	.640	.655	3/4	7/8

TABLE II. Type I, classes 2 and 3, style B, multi-size tool style, tool holders

Tool holder number		Size range of square rectangular, and diameter of cylindrical tool bits, or boring bars	Thickness shank			Depth of shank			Overall length of shank			Length of shank	Height of broached hole		Size of broached hole		Base width of holder head		Width of vise clamp jaw chamber		Wall thickness of tool head		Depth of V in vise clamp jaw	
Type I			dimension A			dimension B			dimension C			dimension D	dimension E		dimension F		dimension G		dimension H		dimension I		dimension J	
Class 2 Style B	Class 3 Style B	Inch	Inches			Inches			Inches			Inches	Inches		Inches		Inches		Inches		Inch		Inch	
		Nominal	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
A-L-40	A-R-50	1/8, 3/16, 1/4, 5/16	3/8	0.35	0.39	7/8	0.85	0.88	5-1/2	5	6	3-1/4	0.29	0.33	0.325	0.331	1/2	5/8	0.302	0.322	0.100	0.150	0.068	0.088
A-L-41	A-R-51	3/16, 1/4, 5/16, 3/8	1/2	.48	.53	1-1/8	1.08	1.13	6-1/2	6	7	3-3/4	.48	.52	.387	.393	5/8	3/4	.365	.385	.125	.187	.099	.119
A-L-42	A-R-52	1/4, 5/16, 3/8, 7/16, 1/2	5/8	.60	.66	1-1/2	1.40	1.51	8	7-1/2	8-1/2	4-3/4	.60	.64	.512	.518	3/4	7/8	.490	.510	.156	.218	.154	.174
A-L-44	A-R-54	5/16, 3/8, 7/16, 1/2, 5/8	7/8	.85	.92	1-7/8	1.75	1.88	9-1/2	9	10	5-1/4	.66	.70	.637	.643	1	1-1/8	.615	.635	.219	.281	.193	.213
A-L-46	A-R-56	1/2, 9/16, 5/8, 11/16, 3/4, 7/8, 1	1-1/4	1.22	1.26	2-1/4	2.22	2.28	13	12-1/2	13-1/2	8-3/4	.98	1.02	1.012	1.018	1-1/2	1-3/4	.990	1.010	.312	.406	.281	.343

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TABLE III. Type II, class 1, and style A of classes 2 and 3, rigid style tool holders

Tool holder number			Size of square tool bit	Thickness of shank			Depth of shank			Overall length of holder			Length of shank	Height of tool head from base of broached square hole		Width of holder head		Height of broached square hole at tool head face end		Size of broached square hole	
Type II				dimension A			dimension B			dimension C			dimension D	dimension E		dimension F		dimension G		dimension H	
Class 1	Class 2 Style A	Class 3 Style A	Inch	Inches			Inches			Inches			Inches	Inches		Inches		Inches		Inch	
			Nom.	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
B-S- 4	B-L- 5	B-R- 6	3/16	5/14	0.30	0.33	1/2	0.48	0.53	4	3-1/2	4-1/2	2-1/2	1/2	3/4	5/16	1/2	3/8	13/32	0.195	0.212
B-S-10	B-L-20	B-R-30	1/4	3/8	.35	.39	7/8	.85	.88	5	4-3/4	5-1/2	3-1/4	3/4	7/8	17/32	39/64	15/32	9/16	.260	.275
B-S-11	B-L-21	B-R-31	5/16	1/2	.48	.53	1-1/8	1.08	1.135	6	5-1/2	6-1/2	3-11/16	31/32	1-1/8	21/32	23/32	19/32	23/32	.327	.338
B-S-12	B-L-22	B-R-32	3/8	5/8	.60	.66	1-3/8	1.27	1.385	7	6-1/2	7-1/2	4-3/4	1-3/32	1-1/4	25/32	55/64	11/16	13/16	.390	.405
B-S-13	B-L-23	B-R-33	7/16	3/4	.73	.79	1-5/8	1.52	1.635	8	7-1/2	8-1/2	5-1/4	1-5/16	1-7/16	15/16	1	25/32	29/32	.453	.467
B-S-14	B-L-24	B-R-34	1/2	7/8	.85	.92	1-3/4	1.62	1.78	9	8-1/2	9-1/2	5-3/4	1-1/2	1-11/16	1-1/16	1-11/64	7/8	1-3/32	.515	.530
B-S-15	B-L-25	B-R-35	5/8	1	.97	1.05	2	1.87	2.01	11	10-1/2	11-1/2	7	1-3/4	2	1-11/32	1-37/64	1-1/16	1-7/32	.640	.660

TABLE IV. Type II, classes 2 and 3, style B, multi-size tool style tool holders

Tool holder number		Size range of square rectangular, and diameter of cylindrical tool bits	Thickness of shank			Depth of shank			Overall length of holder			Length of shank	Height of broached square hole at tool head face end		Size broached hole		Base width of holder head		Width of vise clamp jaw chamber		Wall thickness of tool head		Depth of V in vise clamp jaw	
Type II			dimension A			dimension B			dimension C			dimension D	dimension E		dimension F		dimension G		dimension H		dimension I		dimension J	
Class 2 Style B	Class 3 Style B	Inch	Inches			Inches			Inches			Inches	Inches		Inches		Inches		Inches		Inch		Inch	
		Nominal	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
B-L-40	B-R-50	1/8, 3/16, 1/4, 5/16	3/8	0.35	0.39	7/8	0.85	0.88	5-1/2	5	6	3-1/4	0.35	0.39	0.325	0.331	1/2	5/8	0.302	0.322	0.100	0.150	0.068	0.088
B-L-41	B-R-51	3/16, 1/4, 5/16, 3/8	1/2	.48	.53	1-1/8	1.08	1.13	6-1/2	6	7	3-3/4	.54	.58	.387	.393	5/8	3/4	.365	.385	.125	.187	.099	.119
B-L-42	B-R-52	1/4, 5/16, 3/8, 7/16, 1/2	5/8	.60	.66	1-1/2	1.40	1.51	8	7-1/2	8-1/2	4-3/4	.73	.77	.512	.518	3/4	7/8	.490	.510	.156	.218	.154	.174
B-L-44	B-R-54	5/16, 3/8, 7/16, 1/2, 5/8	7/8	.85	.92	1-7/8	1.75	1.88	9-1/2	9	10	5-1/4	.85	.89	.637	.643	1	1-1/8	.615	.635	.219	.281	.193	.213
B-L-46	B-R-56	1/2, 9/16, 5/8, 11/16, 3/4, 7/8, 1	1-1/4	1.22	1.26	2-1/4	2.22	2.28	13	12-1/2	13-1/2	8-3/4	1.35	1.39	1.012	1.018	1-1/2	1-3/4	.990	1.010	.312	.406	.281	.343

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TABLE V. Type III, classes 1, 2, and 3 styles A and B cut-off tool holders

Tool holder number						Size of cut-off blade	Thickness of shank			Depth of shank			Overall length of tool holder			Length of shank	Overall height of holder		Overall width of holder head		Height of dovetail blade groove	
Type III							dimension A			dimension B			dimension[1] C			dimension D	dimension[1] E		dimension F		dimension[2] G	
Class 1 Style A	Class 2 Style A	Class 3 Style A	Class 1 Style B	Class 2 Style B	Class 3 Style B	Inches	Inches			Inches			Inches			Inches	Inches		Inches		Inch	
						Nominal	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.
C-S- 4	C-L- 5	C-R- 6	C-S- 7	C-L- 8	C-R- 9	1/16 x 3/8	5/16	0.30	0.33	1/2	0.48	0.53	4	3-1/2	4-1/2	2-1/2	1-3/8	1-21/32	5/16	3/8	1/16	3/32
C-S-10	C-L-20	C-R-30	C-S-40	C-L-50	C-R-60	3/32 x 5/8	3/8	.35	.39	7/8	.85	1.00	5	4-1/2	5-1/2	3-1/4	1-7/16	1-11/16	3/8	9/16	3/32	3/16
C-S-11	C-L-21	C-R-31	C-S-41	C-L-51	C-R-61	1/8 x 3/4	1/2	.48	.53	1-1/8	1.08	1.22	6	5-1/2	6-1/2	3-3/4	1-5/8	1-15/16	1/2	23/32	3/16	9/32
C-S-12	C-L-22	C-R-32	C-S-42	C-L-52	C-R-62	1/8 x 7/8	5/8	.60	.66	1-3/8	1.27	1.38	7	6-1/2	7-1/2	4	2-1/16	2-7/32	5/8	7/8	3/16	11/32
C-S-13	C-L-23	C-R-33	C-S-43	C-L-53	C-R-63	3/16 x 1	3/4	.73	.79	1-5/8	1.52	1.63	8	7-1/2	8-1/2	4-1/2	2-5/16	2-1/2	3/4	1	5/16	29/64
C-S-14	C-L-24	C-R-34	------	------	------	3/16 x 1-1/8	7/8	.85	.92	1-3/4	1.62	1.78	9	8-1/2	9-1/2	5-1/2	2-9/16	2-11/16	7/8	1-5/32	5/16	7/16
C-S-15	C-L-24	C-R-35	------	------	------	1/4 x 1-1/4	1	.97	1.05	2	1.87	2.01	11	10	11-1/2	6-1/2	2-3/4	2-7/8	1	1-5/16	5/16	7/16

1/ Not applicable to style B holders.

2/ Measured from bottom surface of tool shank to bottom outside edge of the dovetail style groove.

TABLE VI. Type IV, class 1, rigid threading tool holder

Tool holder number	Thickness of shank			Depth of of shank			Length of shank	Length of offset portion of tool head		Overall height of holder		Overall width of holder head		Center height of clamping screw hole		Thickness of threading tool		Radial length (initial) of threading tool
Type IV	dimension A			dimension B			dimension C	dimension D		dimension E		dimension F		dimension G		dimension H		dimension I
Class 1	Inch			Inches			Inches	Inches		Inches		Inch		Inches		Inch		Inches
	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.
E-L-0 ----------	3/8	0.35	0.39	7/8	0.85	0.88	3-1/8	1-1/8	1-1/4	1-1/16	1-3/8	11/32	7/16	23/32	27/32	0.180	0.253	13/16
E-L-1 ----------	1/2	.48	.53	1-1/8	1.08	1.14	3-3/8	1-1/4	1-5/8	1-7/16	1-9/16	13/32	9/16	7/8	1-1/32	.243	.320	1
E-L-2 ----------	5/8	.60	.66	1-3/8	1.27	1.38	4	1-1/2	1-5/8	1-11/16	1-15/16	35/64	5/8	1-1/8	1-1/4	.352	.383	1-7/32
E-L-3 ----------	3/4	.73	.79	1-5/8	1.52	1.63	5	1-5/8	1-7/8	2	2-1/8	5/8	3/4	1-9/32	1-3/8	.414	.440	1-1/4
E-L-4 ----------	7/8	.85	.92	1-3/4	1.62	1.78	5-3/4	1-5/8	1-7/8	2-1/16	2-3/16	11/16	7/8	1-15/32	1-17/32	.414	.440	1-1/4

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TABLE VII. Type IV, class 2, spring style, threading tool holders

Tool holder number	Size of tool bit	Thickness of shank (A)			Depth of shank (B)			Length of tool holder (C)			Height (D) cutter point to base of tool holder
Type IV	Inch	Inch			Inches			Inches			
Class 2		Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Nominal
E-S-0 ------	1/4	3/8	0.35	0.39	7/8	0.85	0.88	5-1/2	5	6	5/8
E-S-1 ------	5/16	1/2	.48	.53	1-1/8	1.08	1.13	6-1/2	6	7	3/4
E-S-2 ------	3/8	5/8	.60	.66	1-3/8	1.27	1.38	7-1/2	7	8	1-1/16
E-S-3 ------	7/16	3/4	.73	.79	1-5/8	1.62	1.78	8-1/2	8	9	1-3/8

TABLE VIII. Knurls for type V, classes 1 and 2, knurling tool holders

Tool holder number		Thickness of knurls dimension J 1/			Diameter of knurls dimension K			Hold diameter of knurls dimension M	
Type V		Inch			Inches			Inch	
Class 1	Class 2	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Max.
E-S-10	F-R-20	5/16	0.308	0.312	5/8	0.620	0.625	0.218	0.224
E-S-11	F-R-21	3/8	.370	.374	3/4	.745	.750	.250	.255
E-S-12	F-R-22	----	-----	-----	----	-----	-----	-----	-----
E-S-13	------	3/8	.370	.374	1	.985	1.015	.312	.314
E-S-4	------	5/16	.308	.312	5/8	.620	.625	.218	.224

1/ Knurls shall be parallel in thickness within plus or minus 0.003 inch.

TABLE IX. Type V, class 1, single-knurling tool holder

Tool holder number	Thickness of shank dimension A			Depth of shank dimension B			Length of holder dimension C		Length of shank dimension D	Overall height of holder dimension E		Overall width of holder head dimension F	
Type V	Inch			Inches			Inches		Inches	Inches		Inches	
Class 1	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Max.	Min.	Min.	Max.	Min.	Max.
E-S- 4	5/16	0.30	0.33	3/4	0.73	0.85	4-3/8	4-3/4	2-15/16	1-19/16	1-3/4	9/16	21/32
E-S-10	3/8	.35	.39	7/8	.85	.88	4-1/2	4-3/4	3-1/4	1-5/8	1-3/4	19/32	21/32
E-S-11	1/2	.48	.53	1-1/8	1.08	1.13	5-3/4	6	3-3/4	2-1/16	2-3/16	23/32	25/32
E-S-12	5/8	.60	.66	1-3/8	1.27	1.38	6-9/16	6-13/16	5	2-1/16	2-3/16	23/32	25/32
E-S-13	7/8	.85	.92	1-3/4	1.62	1.78	7-3/4	8	5-3/4	2-13/16	2-15/16	31/32	1-1/32

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TABLE X. Type V, class 2, multiple-knurling tool holder

Tool holder number	Thickness of shank dimension A			Depth of shank dimension B			Length of holder dimension C		Length of shank dimension D	Overall height of holder dimension E		Overall width of holder head dimension F	
Type V	Inch			Inches			Inches		Inches	Inches		Inches	
Class 2	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Max.	Min.	Min.	Max.	Min.	Max.
F-R-20 ---	3/8	0.35	0.39	7/8	0.85	0.88	4-1/2	6-1/2	2-1/2	1-7/8	2-3/8	1-1/32	1-3/32
F-R-21 ---	1/2	.48	.53	1-1/8	1.08	1.13	5-1/2	6-1/2	3-1/8	2-1/8	2-3/8	1-3/16	1-5/16
F-R-22 ---	5/8	.60	.66	1-3/8	1.27	1.38	5-1/2	6-1/2	3-1/8	2-1/8	2-3/8	1-3/16	1-5/16

TABLE XI. Boring tools for type VI, boring tool holders

Tool holder number	Diameter range of boring tool			Length of boring tool			Width of cutting face
Type VI	dimension K			dimension L			dimension J
	Inch			Inches			Inch
Class 1	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.
	1/8	1/8	9/64	4	3-7/8	4-3/16	5/16
G-S-10	1/4	1/4	17/64	5	4-7/8	5-3/16	17/32
	3/16	3/16	13/64	4-1/2	4-3/8	4-11/16	7/16
G-S-11	5/16	5/16	21/64	6	5-7/8	6-3/16	5/8
	1/4	1/4	17/64	5	4-7/8	5-3/16	17/32
G-S-12	3/8	3/8	25/64	7	6-7/8	7-3/16	3/4
	5/16	5/16	21/64	6	5-7/8	6-3/16	5/8
G-S-13	7/16	7/16	29/64	8	7-7/8	8-3/16	7/8

TABLE XII. Plain, sleeve, and end cap boring bars

Number of plain boring bars	Number of sleeve boring bars	Number of end cap boring bars	Diameter of boring bars			Length of boring bars			Size of tool bit channel			Size of square tool bit
			dimension K			dimension L			dimension M			dimension M
			Inches			Inches			Inch			Inch
			Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.
PB-1	SB-1	ECB-1	1/2	0.490	0.515	8	7	9	3/16	0.197	0.212	3/16
PB-2	----	ECB-2	9/16	.552	.577	9	8	10	3/16	.197	.212	3/16
PB-3	SB-3	-----	5/8	.615	.640	10	9-1/8	10-1/2	3/16	.197	.212	3/16
PB-4	SB-4	ECB-4	3/4	.740	.765	11	10	12	1/4	.260	.275	1/4
PB-5	SB-5	ECB-5	15/16	.927	.952	14	13	15	5/16	.327	.338	5/16
PB-6	SB-6	ECB-6	1-1/8	1.115	1.140	16	15	18	3/8	.390	.400	3/8
PB-7	SB-7	-----	1-5/16	1.302	1.327	18	18	20	3/8	.390	.400	3/8
PB-8	----	ECB-8	1-5/16	1.302	1.327	18	17	20	7/16	.453	.467	7/16
PB-9	SB-9	-----	1-1/2	1.490	1.515	23	22	24	7/16	.453	.467	7/16
PB-10	----	ECB-10	1-1/2	1.490	1.515	23	22	26	1/2	.515	.530	1/2
PB-11	SB-11	------	1-13/16	1.802	1.827	28	27	29	1/2	.515	.530	1/2
PB-12	-----	ECB-12	1-13/16	1.802	1.827	28	27	31	5/8	.640	.655	5/8
PB-13	SB-13	------	2-1/4	2.240	2.265	34	33	35	5/8	.640	.655	5/8
PB-14	-----	ECB-14	2-1/4	2.240	2.265	31	30	34	3/4	.770	.785	3/4

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TABLE XIII. Type VI, class 1, style A reversible, shank-mounted boring tool holders

Tool holder number	Thickness of shank			Depth of shank			Length of shank	Length of offset portion of tool head		Width of tool support head		Width of square groove opening		Width of 99° V-groove opening		Overall height of holder yoke		Width of holder yoke		Thickness of holder yoke	
Type VI	dimension A			dimension B			dimension C	dimension D		dimension E		dimension F		dimension G		dimension H		dimension I		dimension J	
	Inch			Inches			Inches	Inches		Inch		Inch		Inch		Inches		Inches		Inches	
Class 1 Style A	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
G-S-10	3/8	0.35	0.39	3/4	0.71	0.76	3-3/8	1-3/8	1-13/16	0.495	0.505	0.258	0.268	0.334	0.354	2	2-7/16	27/32	29/32	5/8	11/16
G-S-11	1/2	.48	.53	1	.96	1.11	3-3/4	1-11/16	2-1/16	.589	.630	.323	.333	.396	.416	2-9/16	2-15/16	1	1-3/32	3/4	13/16
G-S-12	5/8	.60	.66	1-1/4	1.21	1.26	4-3/8	1-7/8	2-3/8	.745	.755	.385	.395	.490	.510	3	3-3/8	1-3/16	1-1/4	7/8	1
G-S-13	3/4	.73	.79	1-1/2	1.40	1.51	4-5/8	2-9/16	2-11/16	.870	.880	.447	.457	.583	.603	3-1/2	3-15/16	1-13/32	1-1/2	1-1/8	1-3/16

TABLE XIV. Type VI, class 1, style B, clamp block, shank-mounted boring tool holder

Tool holder number	Thickness of shank			Depth of shank			Length of shank	Length of tool head and clamp block		Axial centerline distance from bar to shank		Capacity range of boring bars (diameter of hole)	
Type VI	dimension A			dimension B			dimension C	dimension D		dimension E			
	Inch			Inches			Inches	Inches		Inches		Inches	
Class 1 Style B	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.
G-M-20	3/8	0.35	0.39	3/4	0.71	0.76	3-3/4	1-5/8	1-7/8	15/16	1-1/16	1/4	5/8
G-M-21	1/2	.48	.53	1-1/8	1.10	1.13	4-5/8	2-1/8	2-3/8	1-3/16	1-5/16	1/4	3/4
G-M-22	5/8	.60	.66	1-3/8	1.34	1.38	5-1/2	2-5/8	2-7/8	1-7/16	1-9/16	3/8	15/16
G-M-23	3/4	.73	.79	1-5/8	1.58	1.63	6-3/8	3-1/16	3-5/16	1-11/16	1-13/16	1/2	1-1/8

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TABLE XV. Type VI, class 1, style C, split clamp, shank-mounted boring tool holder and bushing

Tool holder number	Thickness of shank			Depth of shank			Length of shank	Length of tool head and clamp block		Diameter of bar hole in tool head		Center height from bar to shank [1]	Adaptable bushing number	Outside diameter of bushing	Inside diameter of bushing [1]	Length of bushing		Width of bushing spring slot [2]
Type VI	dimension A			dimension B			dimension C	dimension D		dimension E		dimension F		dimension E [1]	dimension K	dimension D		dimension S
	Inches			Inches			Inches	Inches		Inches		Inches		Inches	Inches	Inches		Inch
Class 1 Style C	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.			Nom.	Nom.	Min.	Max.	
GM-29	5/16	0.29	0.33	3/4	0.73	0.76	3	1-1/2	1-3/4	0.490	0.505	3/4	None	----	----	----	----	----
GM-30	3/8	.35	.39	7/8	.85	.88	3-1/8	1-1/2	1-3/4	.552	.557	13/16	F-00	9/16	1/2	1-11/16	1-13/16	3/32
GM-31	1/2	.48	.53	1-1/8	1.08	1.13	3-1/2	2	2-1/4	.740	.765	1-1/16	F-10	3/4	1/2	2-1/16	2-3/16	3/32
													F-11	3/4	9/16	2-1/16	2-3/16	3/32
GM-32	5/8	.60	.66	1-3/8	1.27	1.38	4-1/8	2-3/16	2-1/2	.927	.952	1-3/16	F-20	15/16	1/2	2-7/16	2-9/16	1/8
													F-21	15/16	9/16	2-7/16	2-9/16	1/8
													F-22	15/16	3/4	2-7/16	2-9/16	1/8
GM-33	3/4	.73	.79	1-5/8	1.52	1.63	4-3/4	2-3/4	3-1/16	1.115	1.140	1-7/16	F-30	1-1/8	1/2	2-13/16	2-15/16	1/8
													F-31	1-1/8	9/16	2-13/16	2-15/16	1/8
													F-32	1-1/8	3/4	2-13/16	2-15/16	1/8
													F-33	1-1/8	15/16	2-13/16	2-15/16	1/8
GM-34	7/8	.85	.91	1-3/4	1.65	1.76	5-1/4	3-1/4	3-9/16	1.302	1.327	1-19/32	F-40	1-5/16	3/4	3-3/8	3-1/2	5/32
													F-41	1-5/16	15/16	3-3/8	3-1/2	5/32
													F-42	1-5/16	1-1/8	3-3/8	3-1/2	5/32
GM-35	1	.98	1.40	2	1.90	2.01	6	3-1/4	4-1/16	1.490	1.515	1-25/32	F-50	1-1/2	15/16	3-13/16	3-15/16	5/32
													F-51	1-1/2	1-1/8	3-13/16	3-15/16	5/32
													F-52	1-1/2	1-5/16	3-13/16	3-15/16	5/32

[1] Plus or minus 1/16 inch.

[2] Plus 0.015 inch, minus 0.005 inch.

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TABLE XVI. Type VI, class 2, style A, adjustable height, boring tool holder

Tool post number	Diameter of T-slot flange		Thickness of T-slot flange		Diameter of T-slot neck		Length of T-slot neck		Overall height of body		Diameter capacity of boring bars		Span of ring adjustment	
Type VI	dimension[1] A		dimension[1] B		dimension[1] C		dimension[1] D		dimension E		dimension K		dimension L	
	Inches		Inches		Inches		Inches		Inches		Inches		Inches	
Class 2 Style A	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
G-L-40	2-15/16	3-1/16	19/32	21/32	1-9/16	1-11/16	23/32	25/32	6-1/2	6-3/4	1/2	1-5/16	1-7/16	1-15/16
G-L-41	3-7/16	3-9/16	23/32	25/32	1-11/16	1-13/16	25/32	27/32	7-1/2	7-3/4	1/2	1-1/2	1-11/16	2-5/16
G-L-42	4-3/16	4-5/16	27/32	29/32	1-15/16	2-1/16	31/32	1-1/32	9	9-1/4	5/8	1-13/16	2-1/16	2-7/8
G-L-43	4-15/16	5-1/16	1-1/32	1-3/32	2-3/16	2-5/16	1-7/32	1-9/32	11	11-1/4	3/4	2-1/4	2-3/8	3-1/4

[1] T-slot flange and neck are furnished large to allow for fitting to individual machines.

TABLE XVII. Type VI, class 2, style B, fixed height, boring tool holders

Tool holder number	Diameter of T-slot flange		Thickness of T-slot flange		Width of flange flat		Diameter of T-slot neck		Diameter of sleeves		Height of sleeves in assembled position		Diameter of boring bars furnished with tool holder			Swing of lathes for which tool holder is adaptable
Type VI	dimension[1] A		dimension[1] B		dimension[1] C		dimension[1] D		dimension E		dimension F					
	Inches		Inches		Inches		Inches		Inches		Inches		Inches			Inches
Class 2 Style B	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Small	Medium	Large	
G-L-50	1-15/16	2-1/16	11/32	13/32	1-7/32	1-9/32	0.600	0.630	1-15/16	2-1/16	2-5/8	2-3/4	1/8	1/2	3/4	8 to 10
G-L-51	3-1/16	3-3/16	17/32	19/32	2-15/32	2-17/32	.975	1.005	3-1/16	3-3/16	3-13/16	3-15/16	1/2	3/4	1-1/8	12 to 16
G-L-52	3-7/16	3-9/16	21/32	23/32	2-31/32	3-1/32	1.225	1.255	3-7/16	3-9/16	4-7/16	4-9/16	5/8	15/16	1-5/16	16 to 18
G-L-53	4-3/16	4-5/16	25/32	27/32	3-15/32	3-17/32	1.225	1.255	4-3/16	4-5/16	5-1/16	5-3/16	3/4	1-1/8	1-1/2	20 to 22
G-L-54	4-13/16	4-15/16	29/32	31/32	3-31/32	4-1/32	1.475	1.505	4-13/16	4-15/16	5-11/16	5-13/16	15/16	1-5/16	1-13/16	24 to 32

[1] T-slot flange and neck are furnished large to allow for fitting to individual machines.

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TABLE XVIII. Type VII, adjustable planer and shaper tool holder

Tool holder number	Thickness of shank dimension A Inches			Depth of shank dimension B Inches			Length of shank dimension C Inches	Diameter of tool head dimension D Inches		Depth of tool head dimension E Inches		Diameter of stud head dimension F Inches		Width of notched tool channel dimension G Inch	
Type VII	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
H-A-4	3/8	0.35	0.39	7/8	0.85	0.88	5-1/2	1-5/16	1-11/16	1-1/32	1-3/16	1-3/8	1-1/2	0.327	0.351
H-A-0	5/8	.60	.66	1-1/4	1.21	1.26	6-1/8	1-7/16	1-11/16	1-5/32	1-1/4	1-1/2	1-5/8	.327	.351
H-A-1	3/4	.73	.79	1-1/2	1.40	1.51	7-9/16	1-11/16	1-15/16	1-5/16	1-7/16	1-3/4	1-7/8	.390	.427
H-A-2	1-1/8	.87	1.05	1-3/4	1.65	1.76	10	2-1/16	2-5/16	1-9/16	1-11/16	2-1/8	2-1/4	.515	.543
H-A-3	1-3/8	1.37	1.43	2	1.95	2.03	12-1/2	2-9/16	2-13/16	1-3/4	1-7/8	2-5/8	2-3/4	.640	.678

TABLE XIX. Type VIII, extension bar shaper tool holder and bushing

Tool holder number	Thickness of shank dimension A Inch			Depth of shank dimension B Inches			Length of shank dimension C Inches	Diamster of tool head dimension D Inches		Diameter of bar hole in tool head dimension E Inches		Depth of tool head dimension F Inches		Capacity range of extension bars dimension K Inches		Bushing number	Outside step diameter of bushing dimension D[1] Inches	Outside step diameter of bushing dimension E[1] Inches	Inside diameter of bushing dimension K[1] Inch	Length of bushing dimension F[1] Inches		Width of bushing spring slot dimension S Inch	
Type VIII	Nom.	Min.	Max.	Nom.	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.		Nom.	Nom.	Nom.	Min.	Max.	Min.	Max.
J-S-3	3/8	0.35	0.38	7/8	0.85	0.88	4	1-3/16	1-5/16	0.50	0.515	1-5/8	1-3/4	1/2	3/4	H-00	15/16	3/4	1/2	1-3/4	1-7/8	3/32	7/64
																H-01	15/16	3/4	5/8	1-3/4	1-7/8	3/32	7/64
J-S-0	1/2	.48	.53	1-1/8	1.08	1.13	5	1-3/16	1-5/16	.740	.765	1-5/8	1-3/4	1/2	3/4	H-00	15/16	3/4	1/2	1-3/4	1-7/8	3/32	7/64
																H-01	15/16	3/4	5/8	1-3/4	1-7/8	3/32	7/64
J-S-1	5/8	.60	.66	1-3/8	1.27	1.38	5-3/4	1-3/8	1-1/2	.927	.952	2	2-1/4	1/2	15/16	H-10	1-3/16	15/16	1/2	2-3/16	2-5/16	3/32	7/64
																H-11	1-3/16	15/16	5/8	2-3/16	2-5/16	3/32	7/64
																H-12	1-3/16	15/16	3/4	2-3/16	2-5/16	3/32	7/64
J-S-2	3/4	.73	.79	1-5/8	1.52	1.63	7	1-11/16	1-13/16	1.115	1.140	2-3/8	2-9/16	1/2	1-1/8	H-20	1-3/8	1-1/8	1/2	2-5/8	2-3/4	1/8	9/64
																H-21	1-3/8	1-1/8	5/8	2-5/8	2-3/4	1/8	9/64
																H-22	1-3/8	1-1/8	3/4	2-5/8	2-3/4	1/8	9/64
																H-23	1-3/8	1-1/8	15/16	2-5/8	2-3/4	1/8	9/64

[1] See figure 28.

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TABLE XX. Extension bars for type VIII shaper tool holders[1/]

Number of extension bar	Diameter of extension bar dimension K, Inches			Length of extension bar dimension L, Inches			Size of square tool-bit channel dimension M, Inch		
	Nom.	Min.	Max.	Nom.	Min.	Max.	Nom.	Min.	Max.
E-B-1	1/2	0.490	0.515	7-1/2	7	8	3/16	0.197	0.219
E-B-2	5/8	.615	.640	8-1/2	8	9	1/4	.260	.281
E-B-3	3/4	.740	.765	10	9-1/2	10-1/2	5/16	.327	.344
E-B-4	15/16	.927	.952	12	11-1/2	12-1/2	3/8	.390	.406
E-B-5	1-1/8	1.115	1.140	14	13-1/2	14-1/2	7/16	.453	.469

[1/] See figure 28.

TABLE XXI. Classification of defects

Categories	Defects
Critical	None defined
Major (AQL = 2.5 percent defective)	
101	Lot not as specified.
102	Manufacturer's type number designation not as ordered.
103	Any characteristics not in accordance with specified requirements (type, class, style); not as specified.
104	Any defect affecting serviceability.
105	Wrenches not supplied as specified.
106	Finish not as specified.
107	Dimensions and tolerances not as specified for the type, class and style.
Minor (AQL = 4.0 percent defective)	
201	Marking not as specified.

TABLE XXII. Classification of defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Not as specified.
	Any component missing or damaged.
Contents (Interior)	Number per container not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container, loose strapping or distortion of container.

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